

(56)

References Cited

U.S. PATENT DOCUMENTS

4,888,057	A	12/1989	Nguyen	
5,604,024	A	2/1997	von Bonin	
5,709,821	A *	1/1998	von Bonin	 C08K 3/016 156/99
8,206,620	B1	6/2012	Bolton	
11,274,439	B2 *	3/2022	Hughes	 E04B 9/16
2002/0071867	A1	6/2002	Gebhard	
2002/0162298	A1	11/2002	Churchill	
2003/0191203	A1 *	10/2003	Oohara	 C08J 9/0019 521/79
2004/0180998	A1	9/2004	Gonzales	
2010/0119858	A1 *	5/2010	Benkhoff	 C09D 15/00 427/160
2010/0146887	A1	6/2010	Wiker	
2010/0223878	A1	9/2010	Lipka	
2011/0023748	A1	2/2011	Wagh	
2014/0076587	A1	3/2014	Herdman	
2017/0121982	A1	5/2017	Bischel	

FOREIGN PATENT DOCUMENTS

WO	WO 2016/173743	A1	11/2016
WO	WO 2017/109174	A1	6/2017

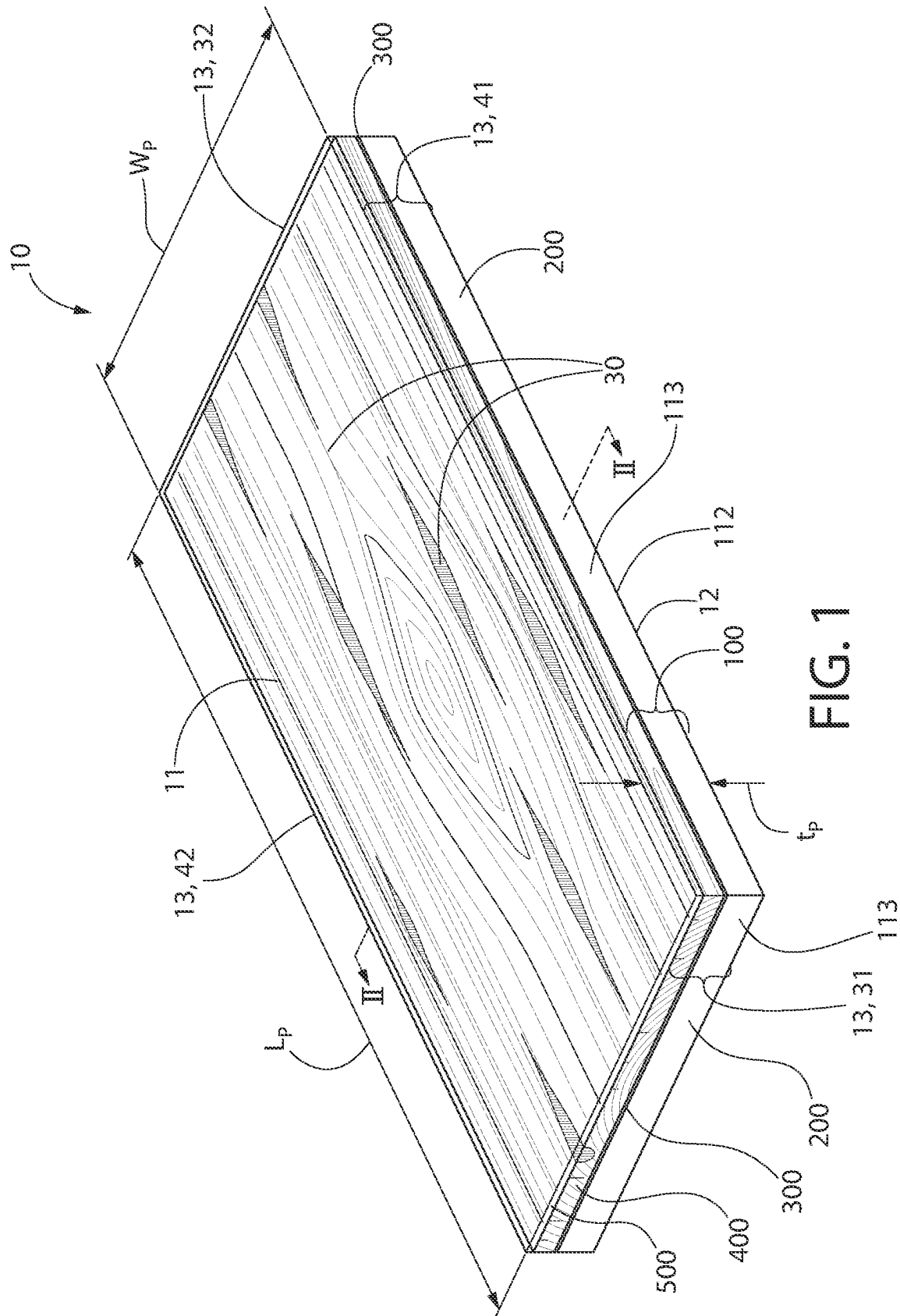
OTHER PUBLICATIONS

Winandy et al., "Effects of a New GUP-B Fire Retardant on Mechanical Properties of Korean Pine When Exposed to Elevated Temperature," Dec. 2005, Forest Products Journal, vol. 55, No. 12, p. 214-220 (2005).

Sidley Chemical Co., Ltd., "Hydroxyethyl Cellulose in Water-Based Paints," Sep. 27, 2013, accessed online Apr. 22, 2021 at www.celluloseether.com/application-hec-water-based-paints (2013).

Shen, "Chapter 11: Review of Recent Advances on the Use of Boron-Based Flame Retardants," 2014, Polymer Green Flame Retardants, Elsevier, p. 367-388. Accessed online Dec. 29, 2020 at www.sciencedirect.com/science/article/pii/B9780444538086000111 (2014).

* cited by examiner



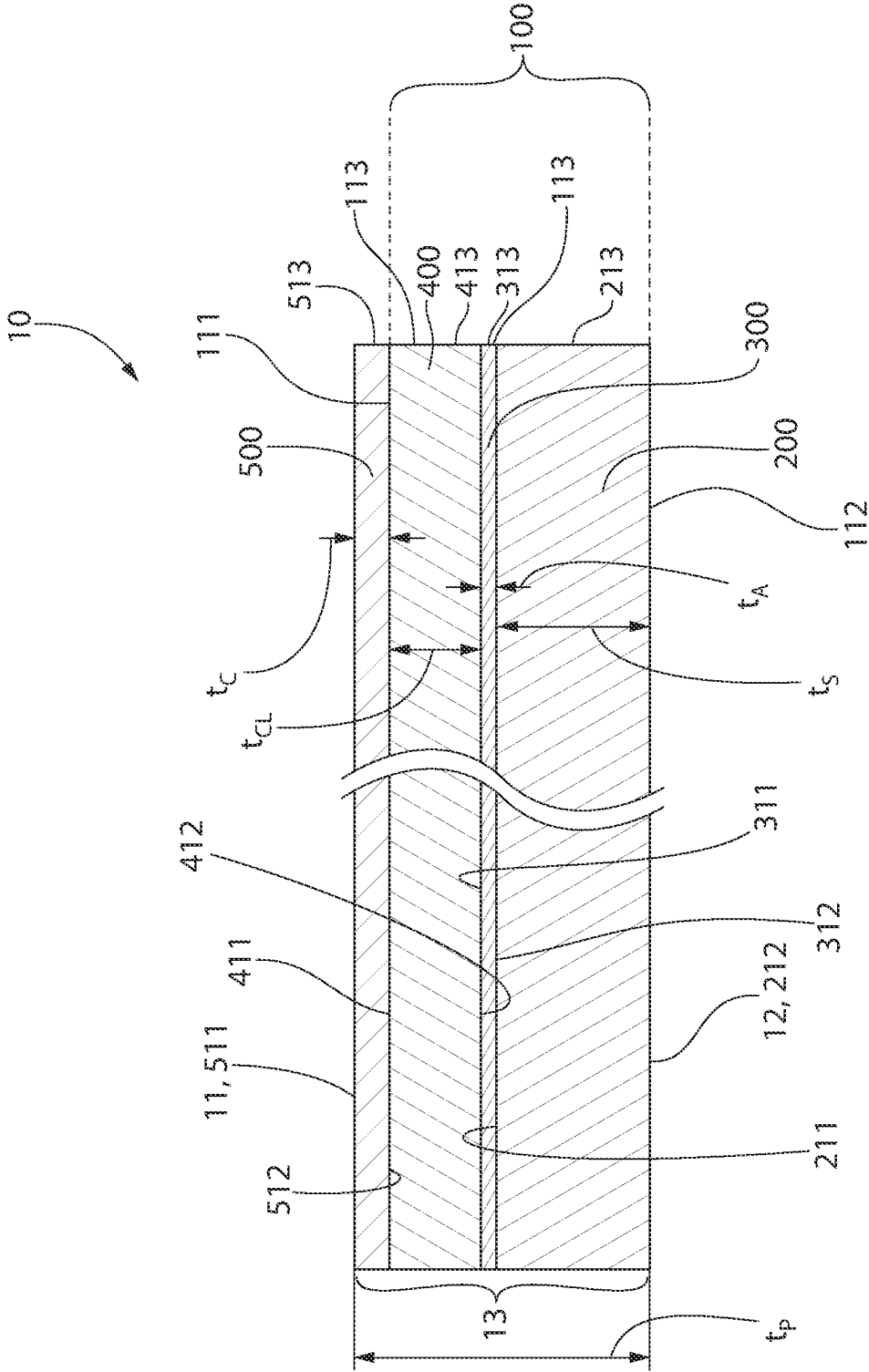


FIG. 2

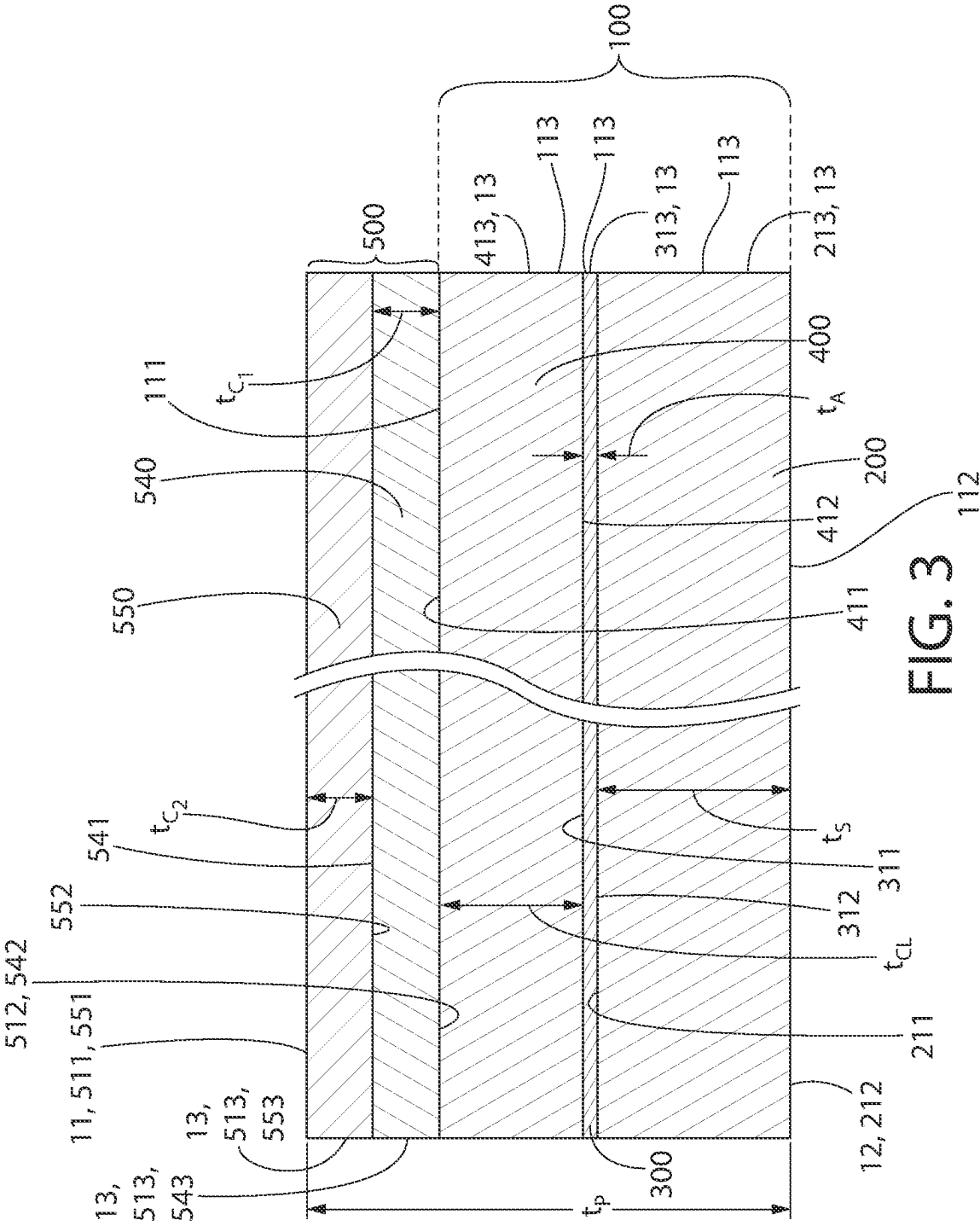
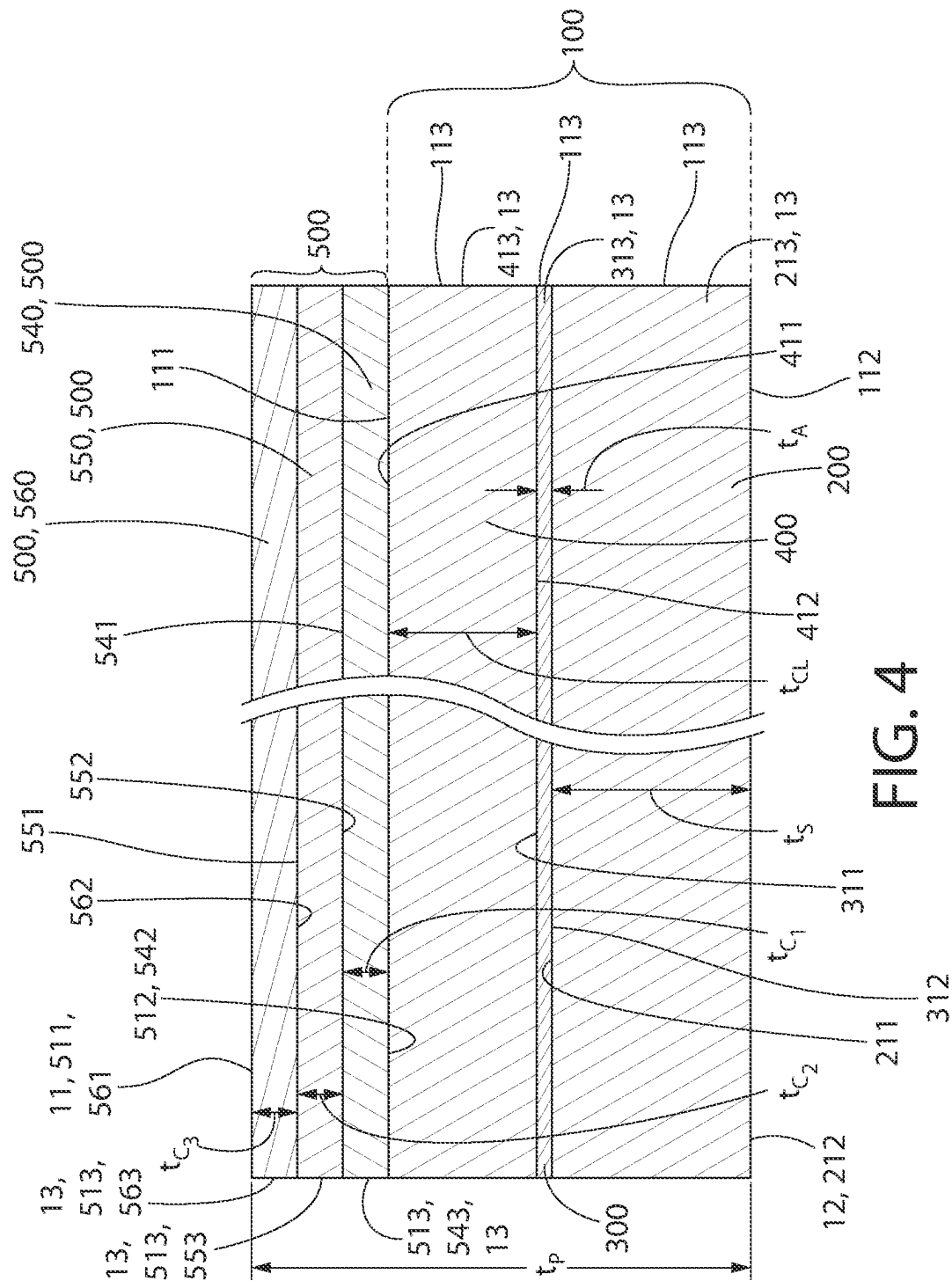


FIG. 3



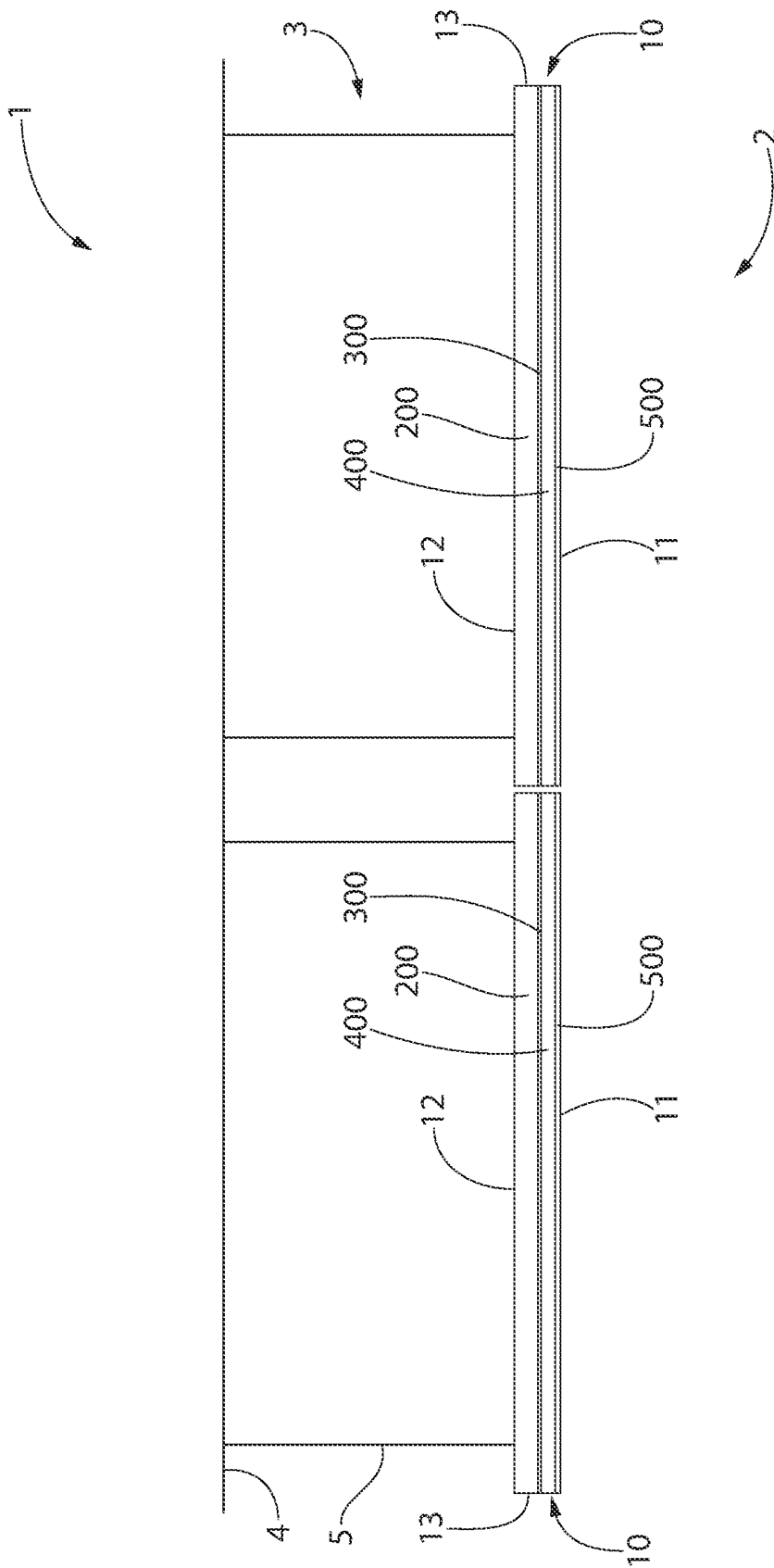
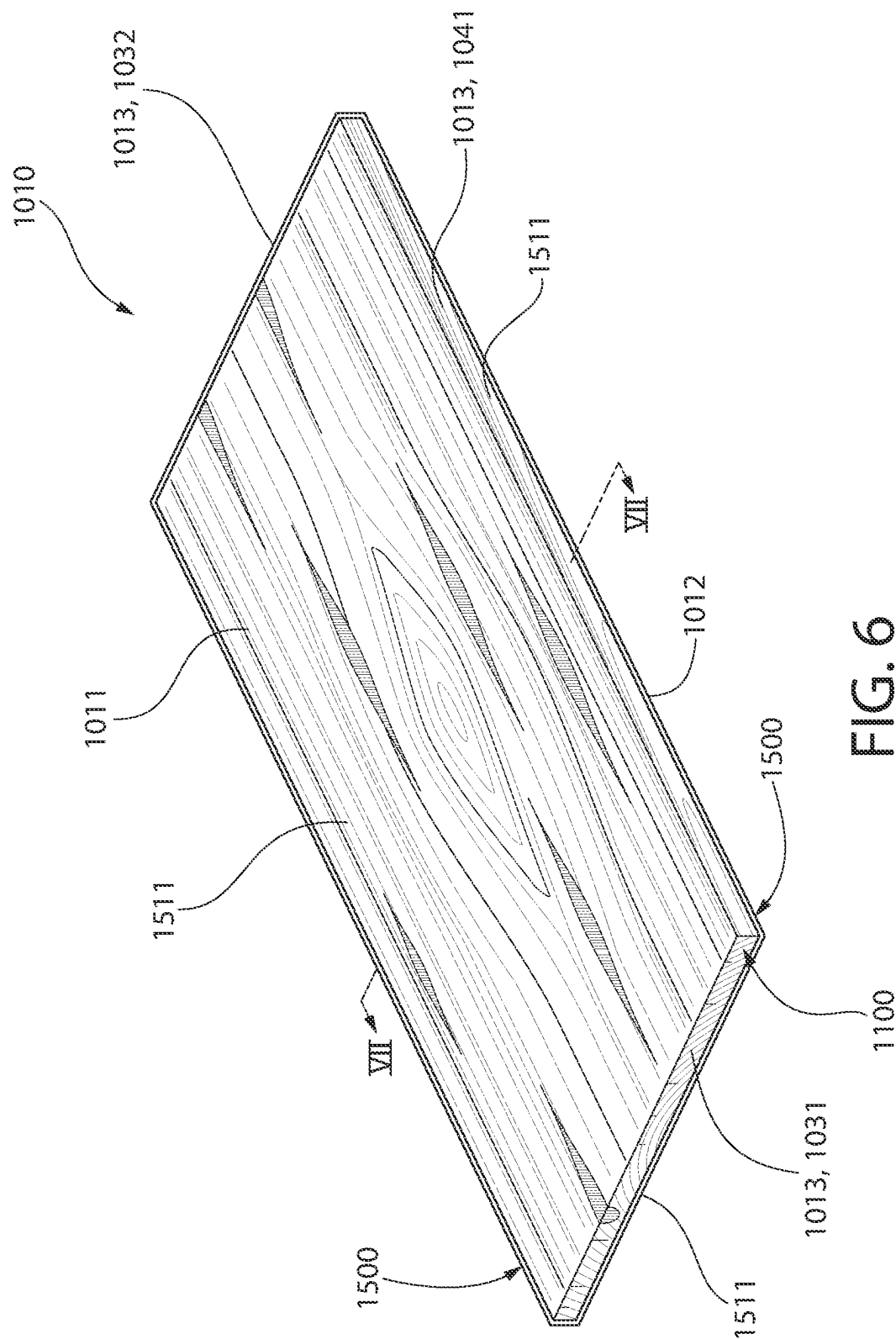


FIG. 5



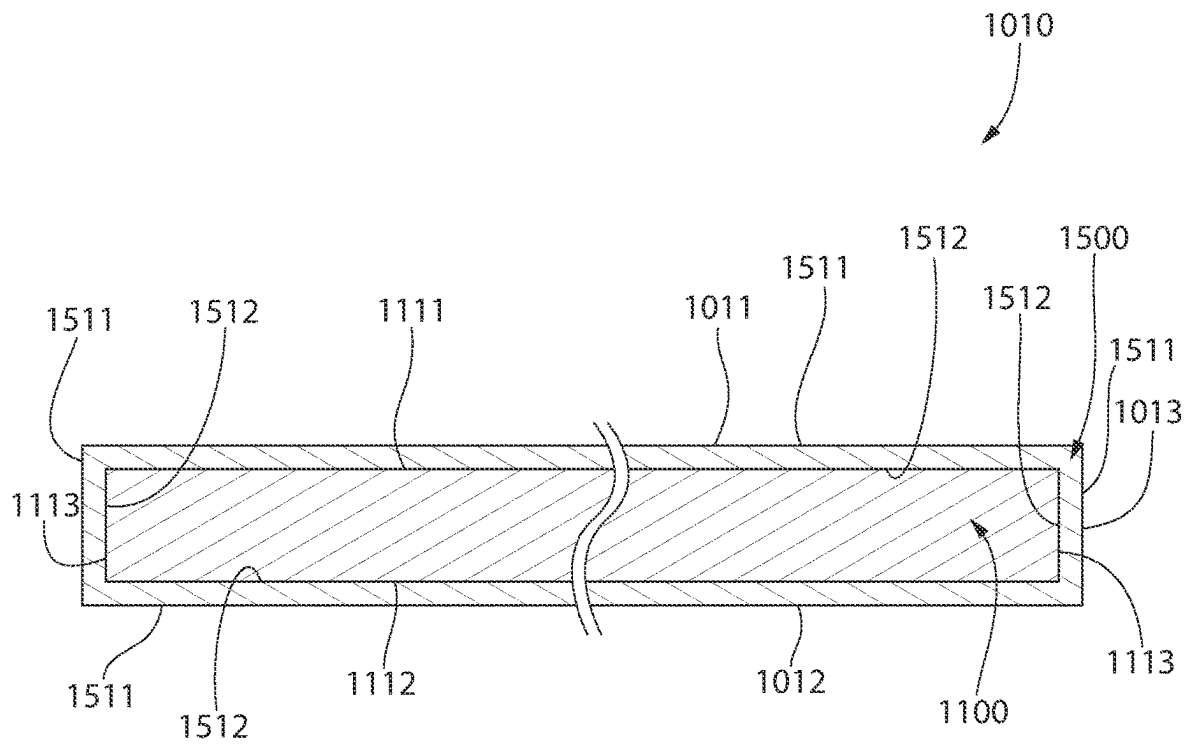
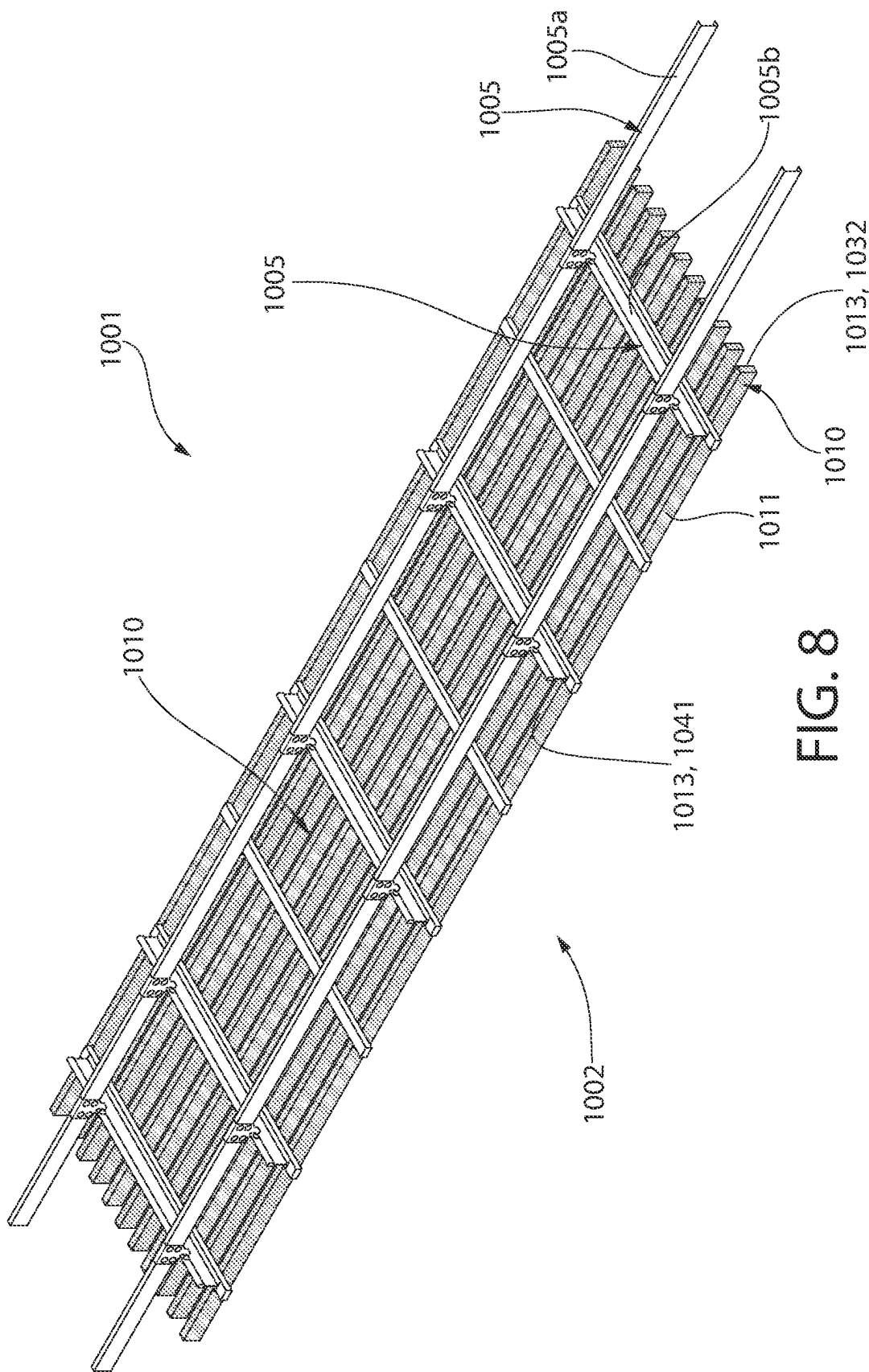


FIG. 7



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FLAME RETARDANT CLEAR COATINGS FOR BUILDING PANELS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 15/994,160, filed May 31, 2018, which claims the benefit of U.S. Provisional Application No. 62/513,115, filed on May 31, 2017. The disclosures of the above applications are incorporated herein by reference.

BACKGROUND

Building products for interior room environments balance interests with respect to cosmetic value, material cost, structural integrity, and fire safety. Previously, maximizing one or two of the aforementioned interests required sacrificing the remaining interests. For example, a building panel that uses natural materials (e.g., natural grain from real wood, as compared to replica grain from printed wood texture) may have superior cosmetic value. However, such building panels also previously had associated safety concerns as either the entire building panel would be made from wood, thereby increasing flammability concerns, thereby endangering individuals by further fueling a fire.

Previous attempts at imparting fire repellency involved applied flame retardant compositions to the major surfaces of the cellulosic materials. However, such previous attempts resulted in degradation of the aesthetic value of the cellulosic substrate due to the flame retardant compositions interfering with the appearance of the resulting building product. Thus, there is a need for building panels that can be formed from natural materials and exhibit superior cosmetic value without degradation the natural aesthetic appearance imparted by such materials.

BRIEF SUMMARY

The present invention is directed to a flame-retardant building panel comprising a body having a first major surface opposite a second major surface, an inorganic coating atop the first major surface, the inorganic coating being optically transparent and flame retardant and comprising a blend of a borate compound and a phosphate compound; and wherein the inorganic flame retardant coating is substantially transparent such that the first major surface of the body is visible through the inorganic coating.

Other embodiments of the present invention include a flame-retardant building panel comprising a body having a first major surface opposite a second major surface, an inorganic coating applied to the first major surface, the inorganic coating formed from a composition comprising a blend of a borate compound and a phosphate compound; and wherein the composition has a maximum pH of 5.

In other embodiments, the present invention is directed to a flame-retardant coating composition comprising a liquid carrier, an inorganic blend comprising a borate compound and a phosphate compound, and wherein the coating composition has a maximum pH of 5 and comprises up to about 2.5 wt. % of organic components based on the total weight of the coating composition.

Other embodiments of the present invention include a flame-retardant building panel comprising a body having a first major surface opposite a second major surface, a top coat comprising a first sub-layer and a second sub-layer, the first sub-layer atop the first major surface of the body and the

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second sub-layer atop the first sub-layer, and wherein the first sub-layer is an organic coating and the second sub-layer is an inorganic coating comprising a blend of a borate compound and a phosphate compound.

Other embodiments of the present invention include a method of forming a flame-retardant building panel comprising a) applying an inorganic coating composition to a major surface of a body, b) drying the inorganic coating composition at an elevated temperature ranging from about 200° F. to about 300° F. to form a flame-retardant coating atop the body; and wherein the inorganic coating composition comprises a liquid carrier, a borate compound and a phosphate compound, and the flame-retardant coating has a solids content of at least 99 wt. % based on the total weight of the flame-retardant coating.

Further areas of applicability of the present invention will become apparent from the detailed description provided hereinafter. It should be understood that the detailed description and specific examples, while indicating the preferred embodiment of the invention, are intended for purposes of illustration only and are not intended to limit the scope of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention will become more fully understood from the detailed description and the accompanying drawings, wherein:

FIG. 1 is top perspective view of the building panel according to the present invention;

FIG. 2 is a cross-sectional view of the building panel according to the present invention, the cross-sectional view being along the II line set forth in FIG. 1;

FIG. 3 is cross-sectional view of a building panel according to other embodiments of the present invention, the cross-sectional view being along the II line set forth in FIG. 1;

FIG. 4 is cross-sectional view of a building panel according to other embodiments of the present invention, the cross-sectional view being along the II line set forth in FIG. 1;

FIG. 5 is a ceiling system comprising the building panel of the present invention;

FIG. 6 is top perspective view of the building panel according to another embodiment of the present invention;

FIG. 7 is a cross-sectional view of the building panel according to the present invention, the cross-sectional view being along the VII line set forth in FIG. 6; and

FIG. 8 is a ceiling system comprising the building panel of FIG. 6.

DETAILED DESCRIPTION

The following description of the preferred embodiment(s) is merely exemplary in nature and is in no way intended to limit the invention, its application, or uses.

As used throughout, ranges are used as shorthand for describing each and every value that is within the range. Any value within the range can be selected as the terminus of the range. In addition, all references cited herein are hereby incorporated by referenced in their entirety. In the event of a conflict in a definition in the present disclosure and that of a cited reference, the present disclosure controls.

Unless otherwise specified, all percentages and amounts expressed herein and elsewhere in the specification should

be understood to refer to percentages by weight. The amounts given are based on the active weight of the material.

The description of illustrative embodiments according to principles of the present invention is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of embodiments of the invention disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. Relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “top,” and “bottom” as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation unless explicitly indicated as such.

Terms such as “attached,” “affixed,” “connected,” “coupled,” “interconnected,” and similar refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. Moreover, the features and benefits of the invention are illustrated by reference to the exemplified embodiments. Accordingly, the invention expressly should not be limited to such exemplary embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features; the scope of the invention being defined by the claims appended hereto.

Unless otherwise specified, all percentages and amounts expressed herein and elsewhere in the specification should be understood to refer to percentages by weight. The amounts given are based on the active weight of the material. According to the present application, the term “about” means $\pm 5\%$ of the reference value. According to the present application, the term “substantially free” less than about 0.1 wt. % based on the total of the referenced value.

Referring to FIGS. 1 and 5, the present invention includes a ceiling system 1 as well as a building panel 10 that may be used in the ceiling system 1. The ceiling system 1 may comprise at least one or more of the building panels 10 installed in an interior space, whereby the interior space comprises a plenum space 3 and an active room environment 2. The plenum space 3 is defined by the space occupied between a structural barrier 4 between floors of a building and the lower major surface 12 of the building panel 10. The plenum space 3 provides space for mechanical lines within a building (e.g., HVAC, electrical lines, plumbing, telecommunications, etc.). The active space 2 is defined by the space occupied beneath the upper major surface 11 of the building panel 10 for one floor in the building. The active space 2 provides room for the building occupants during normal intended use of the building (e.g., in an office building, the active space would be occupied by offices containing computers, lamps, etc.).

Each of the building panels 10 may be supported in the interior space by one or more supports 5. Each of the building panels 10 are installed such that the upper major surface 11 of the building panel 10 faces the active room environment 2 and the lower major surface 12 of the building panel 10 faces the plenum space 3. The building panels 10 of the present invention have superior fire safety performance—particularly when a fire originates in the

active room environment 2—without sacrificing the desired aesthetic appearance of the building panel 10, as discussed herein.

Referring to FIG. 1, the present invention is a building panel 10 having an upper major surface 11, a lower major surface 12 that is opposite the upper major surface 11, and major side surfaces 13 that extend from the upper major surface 11 to the lower major surface 12 to form a perimeter of the building panel 10. The major side surfaces 13 may comprise first and second longitudinal side surfaces 41, 42 extending substantially parallel to each other. The major side surfaces 13 may further comprise first and second transverse side surfaces 31, 32 extending substantially parallel to each other. The first and second longitudinal side surfaces 41, 42 may extend substantially orthogonal to the first and second transverse side surfaces 31, 32.

The building panel 10 may have a panel thickness “ t_p ” as measured from the upper major surface 11 to the lower major surface 12. The panel thickness t_p may range from about 25 mils to about 3,000 mils—including all values and sub-ranges there-between. In some embodiments, the panel thickness t_p may range from about 25 mils to about 600 mils—including all values and sub-ranges there-between. In some embodiments, the panel thickness t_p may range from about 700 mils to about 2,000 mils—including all values and sub-ranges there-between.

The building panel 10 may have a panel length “ L_p ” as measured from the first transverse side surface 31 to the second transverse side surface 32—i.e., the distance along one of the first or second longitudinal side surfaces 41, 42. The panel length L_p may range from about 6 inches to about 100 inches—including all values and sub-ranges there-between. The building panel 10 may have a panel width “ W_p ” as measured from the first longitudinal side surface 41 to the second longitudinal side surface 42—i.e., the distance along one of the first or second transverse side surfaces 31, 32. The panel width W_p may range from about 2 inches to about 60 inches—including all values and sub-ranges there-between. In some embodiments, the panel width W_p may range from about 12 inches to about 60 inches—including all values and sub-ranges there-between.

The building panel 10 of the present invention comprises a body 100 having a coating 500 applied thereto. The body 100 may comprise a first major surface 111 opposite a second major surface 112 and a side surface 113 extending there-between. The body 100 may be formed from a cellulosic material (e.g., wood), metal, organic polymer, inorganic polymer, and combinations thereof. The body 100 may be formed from a single layer of material (also referred to as an integral structure) or the body 100 may have a laminate structure formed from at least two layers. As discussed in greater detail herein, the body 100 having a laminate structure may comprise one or more layers of a cellulosic layer 400, a substrate layer 200, and/or an adhesive layer 300. Although not pictured, the coating 500 of the present invention may be applied to a non-woven scrim. Non-limiting examples of non-woven scrim include fiberglass non-woven scrims. The non-woven scrim may form at least one of the first or second major surface 11, 12 of the building panel 10.

The building panel 10 may comprise a decorative pattern 30 that is visible from the upper major surface 11, the lower major surface 12, and/or the major side surface 13. The decorative pattern 30 may comprise a pattern formed from natural materials, such as cellulosic materials (e.g., wood grain, knots, burl, etc.) or synthetic materials, such as a printed ink. The decorative pattern 30 may be a body decorative pattern that exists on one of the first major surface

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111, second major surface 112, or side surface 113 of the body 100, whereby the body decorative pattern is visible through the coating 500.

The coating 500 may be independently applied to each of the first major surface 111, the second major surface 112, and/or the side surface 113 of the body 100. In a preferred embodiment, the coating 500 is applied to the first major surface 111 of the body 100—as shown in FIG. 2. In another preferred embodiment, the coating 1500 may be applied to each of the first major surface 1111, the second major surface 1112, and the side surface 1113 of the body 1100 such that the coating 1500 encapsulates the body 1100.—as shown in FIGS. 6 and 7.

The coating 500 may be clear or substantially clear. For the purposes of this application, the phrases “substantially clear” or “substantially transparent” refers to materials that have the property of transmitting light in such a way that a normal, human eye (i.e., one belonging to a person with so-called “20/20” vision) or a suitable viewing device can see through the material distinctly. The level of transparency should generally be one which permits a normal, human eye to distinguish objects having length and width on the order of at least 0.5 inches, and should not significantly distort the perceived color of the original object. The coating 500 should be substantially clear (or substantially transparent) such that the underlying body decorative feature can be visible from the upper major surface 11 of the building panel 10 as the decorative pattern 30 on the overall building panel 10, as discussed further herein. The term “substantially clear” or “substantially transparent” may also refer to the coating having at least 70% optical clarity, whereby 100% optical clarity refers to an underlying surface being completely unhindered visually by the coating 500.

Referring now to FIG. 2, the coating 500 may comprise an upper coating surface 511 opposite a lower coating surface 512. The coating 500 may comprise a coating side surface 513 that extends from the upper coating surface 511 to the lower coating surface 512 and forms a perimeter of the coating 500. The coating side surface 513 may form a portion of the major side surface 13 of the building panel 10. Stated otherwise, the major side surface 13 of the building panel 10 may comprise the coating side surface 513. The coating 500 may have a coating thickness “ t_c ” ranging from about 0.5 mils to about 3.0 mils—including all values and sub-ranges there-between—as measured from the upper coating surface 511 to the lower coating surface 512.

The coating 500 may comprise an inorganic composition that is substantially clear. According to the present invention, the phrase “inorganic composition” refers to a dry-state composition having less than 3 wt. % of organic compounds present based on the total dry-weight of the referenced inorganic composition, preferably less than 1.5 wt. % of organic compounds present based on the total dry-weight of the referenced inorganic composition. According to the present invention, the phrase “inorganic composition” may also refer to a wet-state composition that has less than 5.0 wt. % of organic compounds present based on the total wet-weight of the referenced inorganic composition, preferably less than 3.0 wt. % of organic compounds present based on the total wet-weight of the referenced inorganic composition.

The phrase “dry-weight” refers to the weight of a referenced component or composition without the weight of any carrier. Thus, when calculating the amounts of components based on dry-weight, the calculation are to be based solely on the solid components (e.g., binder, filler, hydrophobic component, fibers, etc.) and should exclude any amount of

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residual carrier (e.g., water, VOC solvent) that may still be present from a wet-state, which will be discussed further herein. Additionally, according to the present invention, the phrase “dry-state” refers to a component or composition that is substantially free of carrier, as compared to the phrase “wet-state,” which refers to that component still containing various amounts of carrier. The term “wet-state” refers to a component or composition that further comprises a carrier. Similarly, the phrase “wet-weight” refers to a total weight of component or composition that includes the weight of the carrier when in the wet-state.

The inorganic composition may be a glass-forming composition. According to the present invention, a “glass-forming composition” comprises a first glass component and a second glass component that together react to form an inorganic glass-like surface when exposed to elevated temperatures—such as elevated temperatures during a fire. The inorganic composition of the present invention exhibits a pH ranging from about 1.5 to about 5—including all pHs and sub-ranges there-between. In a preferred embodiment, the pH ranges from about 1.7 to about 4—including all pHs and sub-ranges there-between. In a preferred embodiment, the pH is less than about 3.

The first glass component comprises a phosphate compound. The second glass component comprises a borate compound. The first glass component and the second glass component may be present in a weight ratio ranging from about 5:1 to about 80:1—including all ratios and sub-ranges there-between.

The first glass component may be present in an amount ranging from about 40.0 wt. % to about 80.0 wt. %—including all wt. % and sub-ranges there-between—based on the total weight of the inorganic composition in the dry-state. In a preferred embodiment, the first glass component may be present in an amount ranging from about 55.0 wt. % to about 75.0 wt. %—including all wt. % and sub-ranges there-between—based on the total weight of the inorganic composition in the dry-state. The second glass component may be present in an amount ranging from about 0.5 wt. % to about 15.0 wt. %—including all wt. % and sub-ranges there-between—based on the total weight of the inorganic composition in the dry-state. In a preferred embodiment, the second glass component may be present in an amount ranging from about 1.0 wt. % to about 10.5 wt. %—including all wt. % and sub-ranges there-between—based on the total weight of the inorganic composition in the dry-state.

Non-limiting examples of the phosphate compound of the first glass component include phosphoric acid (H_3PO_4), pyrophosphoric acid ($H_4P_2O_7$), polyphosphoric acid, sodium phosphate, potassium phosphate, aluminum tris (dihydrogen phosphate), other phosphate-ion forming compounds, and combinations thereof. Non-limiting examples of the borate compound of the second glass component include boron trioxide (B_2O_3), zinc borate, and other soluble borate forming compound at pH between 1 and 5, and combinations thereof. Several variants of zinc borate exist and include Zinc borate Firebrake ZB ($2ZnO \cdot 3B_2O_3 \cdot 3.5H_2O$), Zinc borate Firebrake 500 ($2ZnO \cdot 3B_2O_3$), Zinc borate Firebrake 415 ($4ZnO \cdot B_2O_3 \cdot H_2O$), ZB-467 ($4ZnO \cdot 6B_2O_3 \cdot 7H_2O$), and ZB-223 ($2ZnO \cdot 2B_2O_3 \cdot 3H_2O$). According to some embodiments the zinc borate may also serve as a fungicide.

Upon exposure to elevated temperatures, the first and second glass compositions react to form a phosphoborate glass layer (also referred to as the “glass layer”). The glass layer forms a hard protective and heat-insulative barrier that is especially helpful in preventing the body 100 from

igniting at elevated temperatures, for example when the body **100** is formed from a cellulosic material as discussed further herein. The heat-insulative barrier formed by the coating **500** is especially useful when the upper major surface **11**, lower major surface **12**, and/or side surface **13** of the building panel **10** is exposed to heat from a fire that exists in the active room environment **2** of the ceiling system **1** (as shown in FIG. **5**). The heat-insulative barrier created by the inorganic composition slows and prevents further propagation of heat and flame through the coating **500** and, therefore, through the rest of the body **100** of the building panel **10**.

The inorganic composition may optionally comprise other additives or fillers such as, but not limited to fire retarding compounds (also referred to as “flame retardant”), adhesion promoters, viscosity modifying agents, wetting agents, catalyst, cross-linkers, and ultra-violet stabilizers. According to some embodiments, the inorganic composition may further comprise organic compounds so long as the overall inorganic composition includes less than 5 wt. % of organic compounds in the overall inorganic composition. According to some embodiments, the inorganic composition may be substantially free of blowing-agent. The wetting agent may be present in a non-zero amount that is less than about 0.1 wt. %—based on the total dry-weight of the inorganic composition.

The filler may be present in the inorganic coating in an amount ranging from about 15 wt. % to about 75 wt. %—including all amounts and sub-range there-between—based on the total dry weight of the inorganic coating. Non-limiting examples of filler may include calcium carbonate (CaCO_3), aluminum carbonate ($\text{Al}_2(\text{CO}_3)_3$), lithium carbonate (LiCO_3), magnesium carbonate (MgCO_3), fumed silica, aluminum oxide (Al_2O_3), and combinations thereof.

The flame retardants may be present in the coating **500** in an amount ranging from about 0 wt. % to about 50 wt. %—including all values and sub-ranges there-between—based on the total weight of the coating **500**. Non-limiting examples of flame retardant may include ammonium hydroxide, magnesium hydroxide, huntite, hydromagnesite, silica, polyphosphate, chloride salts—such as sodium chloride, antimony oxide, and borates, such as calcium borate, magnesium borate, zinc borate, and combinations thereof.

A non-limiting example of viscosity modifying agent may include hydroxyethyl cellulose, bentonite, polyacrylic rheology modifier, polyurethane rheology modifier, silica, and combination thereof. Hydroxyethyl cellulose is an organic compound—therefore, the hydroxyethyl cellulose may be present in the inorganic composition in an amount that ranges from a non-zero amount to less than 5 wt. % based on the total dry-weight of the inorganic composition. In a preferred embodiment, the hydroxyethyl cellulose may be present in the inorganic composition in an amount that ranges from a non-zero amount to less than 3 wt. % based on the total dry-weight of the inorganic composition.

The inorganic composition may further comprise a cross-linker that facilitates curing of the coating at lower temperatures. Non-limiting example of cross-linker include triethanolamine, polyol, amine or polyamine, as well as other suitable crosslinker that do not inhibit film formation of the coating. In a preferred embodiment, the crosslinker comprises triethanolamine, which is an organic compound. Therefore, the triethanolamine may be present in the inorganic composition in an amount that ranges from a non-zero amount to less than 5 wt. % based on the total wet-weight of the inorganic composition (i.e., before cross-linking). In a preferred embodiment, the triethanolamine may be present

in the inorganic composition such that the total organic content ranges from a non-zero amount to less than 3 wt. % based on the total wet-weight of the inorganic composition.

According to the present invention, the coating **500** may be comprised of a single integral layer (FIG. **2**) or a plurality of sub-layers **540**, **550**, **560** (FIGS. **3** and **4**). The coating **500** show in FIG. **2** having a single integral layer comprises the inorganic composition of the present invention.

Referring now to FIG. **3**, the coating **500** of the present invention may comprise may comprise a first sub-layer **540** and a second sub-layer **550**, whereby the first sub-layer **540** is directly atop one or more of the first major surface **111**, second major surface **112**, and/or side surface **113** of the body **100**. The second sub-layer **550** may be directly atop the first sub-layer **540**. The second sub-layer **550** comprises the inorganic composition of the present invention. In such embodiments, the inorganic composition may be present in an amount ranging from about 95 wt. % to about 100 wt. %—including all values and sub-ranges there-between—based on the total dry-weight of the second sub-layer **550**.

Referring now to FIG. **4**, the coating **500** of the present invention may comprise may comprise a first sub-layer **540**, a second sub-layer **550**, and a third sub-layer **560**, whereby the first sub-layer **540** is directly atop one or more of the first major surface **111**, second major surface **112**, and/or side surface **113** of the body **100**. The second sub-layer **550** may be directly atop the first sub-layer **540**, and the third sub-layer **560** may be directly atop the second sub-layer **550**. The second sub-layer **550** may comprise the inorganic composition of the present invention. In such embodiments, the inorganic composition may be present in an amount ranging from about 95 wt. % to about 100 wt. %—including all values and sub-ranges there-between—based on the total dry-weight of the second sub-layer **550**.

According to the present invention, the first sub-layer **540** may also be referred to as a “base coating.” The first sub-layer **540** may be an organic coating. According to the present invention, the term “organic coating” refers to a coating in the dry-state that comprises at least 10 wt. % of organic compounds based on the total weight of the referenced organic coating in the dry-state, preferably at least 15 wt. % of organic compounds based on the total weight of the referenced organic coating in the dry-state.

The first sub-layer **540** may comprise a polymer binder in an amount ranging from about 70 wt. % to about 95 wt. %—including all values and sub-ranges there-between—based on the total dry-weight of the first sub-layer **540**.

The polymer binder may comprise polymer produced from unsaturated monomers. Specifically, the polymer may be a homopolymer or copolymer produced from ethylenically unsaturated monomers, such as styrene, alpha-methylstyrene, polymethylsiloxane, vinyl toluene, ethylene, propylene, vinyl acetate, vinyl chloride, vinylidene chloride, acrylonitrile, acrylamide, methacrylamide, acrylic acid, methacrylic acid, (meth)acryloxy-propionic acid, itaconic acid, aconitic acid, maleic acid, monomethyl maleate, monomethyl fumarate, monomethyl itaconate, various (C_1 - C_{20}) alkyl or (C_3 - C_{20}) alkenyl esters of (meth)acrylic acid, various lacquers, latex-based binders and the like. The expression (meth)acrylic, as used herein, is intended to serve as a generic expression embracing both acrylic and methacrylic acid and esters thereof e.g., methyl (meth)acrylate, ethyl (meth)acrylate, butyl (meth)acrylate, isobutyl (meth)acrylate, 2-ethyl hexyl(meth)acrylate, benzyl (meth)acrylate, lauryl (meth)acrylate, oleyl (meth)acrylate, palmityl (meth)acrylate, stearyl (meth)acrylate and the like. In other embodiments, the coating polymer binder may include poly-

mer comprising polyurethane, polyester, polyester-modified polyurethane, or a combination thereof.

The polymer binder of the first sub-layer **540** may have a glass transition temperature T_g ranging from about 10° F. to about 115° F.—including all values and sub-ranges there-between.

The first sub-layer **540** may further comprise one or more nonionic surfactants in the amount of 0.1 wt. % to about 1.0 wt. %—based on the total dry-weight of the first sub-layer **540**—including all amount and sub-ranges there-between. The nonionic surfactant component can be a single surfactant or a mixture of two or more nonionic surfactants, the mixture having appropriate HLB values. Suitable nonionic surfactants include but are not limited to ethoxylated nonylphenols, ethoxylated alcohols, ethoxylated castor oil, polyethylene glycol fatty acid esters, and ethyleneglycol-propyleneglycol copolymers. The contemplated nonionic surfactants include ethoxylated nonylphenols and polyethylene glycol fatty acid esters.

The second sub-layer **550** may be an inorganic coating that comprises the inorganic glass-forming composition of the present invention. The second sub-layer **550** may be a combination of multiple coatings of the inorganic glass-forming composition. The second sub-layer **550** may be substantially clear. According to the present invention, the phrase “inorganic coating” refers to the coating having less than 5 wt. % of organic compounds present based on the total dry weight of the referenced inorganic coating, preferably less than 3 wt. % of organic compounds present based on the total weight of the referenced inorganic coating in the dry-state. According to some embodiments, the inorganic coating of the second sub-layer **550** may be formed entirely from the inorganic composition of the present invention.

The third sub-layer **560** may be referred to as “topcoat.” The third sub-layer **560** may be an inorganic coating according to the present invention or an organic coating. The third sub-layer **560** may comprise a sealant composition. The sealant composition may comprise a sealant polymer binder and a flame retardant. Referring now to FIG. 4, other embodiments provide that the coating **500** may further comprise a third sub-layer **560** atop the second sub-layer **550**, which is atop the first sub-layer **540** that is atop the cellulosic layer **400**. The third sub-layer **560** may be formed from a moisture barrier composition that imparts moisture barrier properties to the resulting third sub-layer **560**. The moisture barrier composition may be comprised of hydrophobic polymeric binder, which may or may not be cross-linked, as well as various additives and fillers.

The sealant polymer binder may be present in an amount ranging from about 10 wt. % to about 100 wt. %—including all values and sub-ranges there-between—based on the total weight of the sealant composition in the dry-state. The flame retardant may be present in the cellulosic-layer sealant composition in an amount ranging from about 0 wt. % to about 50 wt. %—including all values and sub-ranges there-between—based on the total dry-weight of the cellulosic-layer sealant composition.

The sealant polymer binder may comprise one or more vinyl or acrylic homopolymers or copolymers formed from ethylenically unsaturated monomers such as ethylene or butadiene and vinyl monomers such as styrene, vinyl esters such as vinyl acetate, vinyl propionate, vinyl butyrate, acrylic acid, methacrylic acid, or esters of acrylic acid and/or esters of methacrylic acid. The esters of acrylic or methacrylic acid may have an alkyl ester portion containing 1 to 12 carbon atoms as well as aromatic derivatives of acrylic and methacrylic acid, and can include, for example, acrylic

and methacrylic acid, methyl acrylate and methyl methacrylate, ethyl acrylate and ethyl methacrylate, butyl acrylate and butyl methacrylate, propyl acrylate and propyl methacrylate, 2-ethyl hexyl acrylate and 2-ethyl hexyl methacrylate, cyclohexyl acrylate and cyclohexyl methacrylate, decyl acrylate and decyl methacrylate, isodecyl acrylate and isodecyl methacrylate, benzyl acrylate and benzyl methacrylate and various reaction products such as butyl, phenyl, and cresyl glycidyl ethers reacted with acrylic and methacrylic acids. In a preferred embodiment, the sealant binder comprises a self-crosslinking acrylic binder. Non-limiting examples of hydrophobic polymeric binder produced from unsaturated monomers. Specifically, the hydrophobic polymer may be a homopolymer or copolymer produced from ethylenically unsaturated monomers, such as styrene, alpha-methylstyrene, vinyl toluene, ethylene, propylene, vinyl acetate, vinyl chloride, vinylidene chloride, acrylonitrile, acrylamide, methacrylamide, acrylic acid, methacrylic acid, (meth)acryloxy-propionic acid, itaconic acid, aconitic acid, maleic acid, monomethyl maleate, monomethyl fumarate, monomethyl itaconate, various (C_1 - C_{20}) alkyl or (C_3 - C_{20}) alkenyl esters of (meth)acrylic acid and the like. The expression (meth)acrylic, as used herein, is intended to serve as a generic expression embracing both acrylic and methacrylic acid and esters thereof e.g., methyl (meth)acrylate, ethyl (meth)acrylate, butyl (meth)acrylate, isobutyl (meth)acrylate, 2-ethyl hexyl(meth)acrylate, benzyl (meth)acrylate, lauryl (meth)acrylate, oleyl (meth)acrylate, palmityl (meth)acrylate, stearyl (meth)acrylate and the like. In other embodiments, the hydrophobic polymeric binder may include polymer comprising polyurethane, polyester, polyester-modified polyurethane, epoxy or a combination thereof.

The hydrophobic polymer may be present in an amount ranging from about 70 wt. % to about 100 wt. %—including all values and sub-ranges there-between—based on the total weight of the moisture barrier composition.

The flame retardant of the first sub-layer **540** may include ammonium hydroxide, magnesium hydroxide, huntite, hydromagnesite, silica, polyphosphate, melamine cyanurate, chloride salts—such as sodium chloride, antimony oxide, and borates, such as calcium borate, magnesium borate, zinc borate, and combinations thereof.

Generally, the coating **500** may be applied directly to one of the first major surface **111**, second major surface **112**, and/or side surface **113** of the body **100**, optionally with the addition of a carrier such as water. The coating **500**—including each sub-layer **540**, **550**, **560**—may be applied by spray, roll-coating, dip coating, curtain coating, brushing, blade coating, or the like, followed by drying and/or curing (optionally with the addition of heat) for a period of time to form the coating **500** atop the cellulosic layer **400**—as discussed in greater detail herein.

The first sub-layer **540** may be applied in the wet-state directly to at least one of the first major surface **111**, second major surface **112**, and/or side surface **113** of the body **100**. In the wet-state, the first sub-layer **540** may comprise a carrier in an amount ranging from about 35 wt. % to about 55 wt. %—including all values and sub-ranges there-between—based on the total weight of the wet-state first sub-layer **540**. In the wet-state, the first sub-layer **540** may be applied in an amount such that the first sub-layer **540** has a wet thickness ranging from about 0.5 mils to about 2.5 mils—including all values and sub-ranges there-between. The carrier may be selected from water, an organic solvent, or a combination thereof. In a preferred embodiment, the wet-state sealant composition is a waterborne system having

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a carrier of water and a low VOC (i.e., volatile organic compound) content—i.e. substantially free of VOC solvents. The first sub-layer **540** in the wet-state may then be cured or dried (optionally with the addition of heat) for a first time period, thereby forming the first sub-layer **540** atop the body **100**.

The resulting first sub-layer **540** may comprise a first sub-layer upper surface **541** and a first sub-layer lower surface **542** opposite the first sub-layer upper surface **541**. The first sub-layer **540** may have a first sub-layer thickness “ t_{C1} ” as measured from the first sub-layer upper surface **541** to the first sub-layer lower surface **542**. The first sub-layer thickness t_{C1} may range from 0.1 mils to 1.0 mils—including all values and sub-ranges there-between. The first sub-layer **540** may comprise a first sub-layer side surface **543** that extends from the first sub-layer upper surface **541** to the first sub-layer lower surface **542** and forms a perimeter of the first sub-layer **540**.

As previously discussed, the body **100** may be formed from a cellulosic material, which comprises pores. Thus, once the first sub-layer **540** is applied to one of the first major surface **111**, second major surface **112**, and/or side surface **113** of the body **100**, at least a portion of the first sub-layer **540** may penetrate into and seal the pores in a direct extending from one of the first major surface **111**, second major surface **112**, and/or side surface **113** of the body **100** toward the center of the body **100**. The first sub-layer **540** in the wet-state may then be dried, optionally, at an elevated temperature, thereby rendering the first sub-layer **540** in the dry state.

The second sub-layer **550** may be formed by directly applying the previously discussed inorganic composition in the wet-state to the first sub-layer upper surface **541** of the first sub-layer **540**. In the wet-state, the second sub-layer **550** may be applied in an amount such that the second sub-layer **550** has a wet thickness ranging from about 0.1 mils to about 1.0 mils—including all values and sub-ranges there-between.

The inorganic composition may then be dried (optionally with the addition of heat) for a second time period of time, thereby forming the second sub-layer **550** atop the first sub-layer **540**. The resulting second sub-layer **550** may comprise a second sub-layer upper surface **551** and a second sub-layer lower surface **552** opposite the second sub-layer upper surface **551**. The second sub-layer **550** in the wet-state may be dried at a temperature ranging from about 190° F. to about 300° F.—including all temperatures and sub-ranges there-between. In a preferred embodiment, the second sub-layer **550** in the wet-state may be dried at a temperature ranging from about 200° F. to about 290° F.—including all temperatures and sub-ranges there-between.

The second sub-layer **550** may have a second sub-layer thickness “ t_{C2} ” as measured from the second sub-layer upper surface **551** to the second sub-layer lower surface **552**. The second sub-layer thickness t_{C2} may range from about 0.02 mils to about 0.7 mils. The second sub-layer **550** may comprise a second sub-layer side surface **553** that extends from the second sub-layer upper surface **551** to the second sub-layer lower surface **552** and forms a perimeter of the second sub-layer **550**.

The first sub-layer side surface **543** and the second sub-layer side surface **553** may form at least a portion of the coating side surface **513**. Stated otherwise, the coating side surface **513** may comprise the first sub-layer side surface **543** and the second sub-layer side surface **553**. The overall

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coating thickness t_C of coating **500** may be the summation of the first sub-layer thickness t_{C1} and the second sub-layer thickness t_{C2} —as follows:

$$t_C = t_{C1} + t_{C2}$$

According to these embodiments, the first sub-layer lower surface **542** of the first sub-layer **540** may contact the upper cellulosic surface **411** of the cellulosic layer **400**. The first sub-layer upper surface **541** may contact the second sub-layer lower surface **552** of the second sub-layer **550**. The second sub-layer upper surface **551** may form at least part of the upper coating surface **511** of the coating **500**. The first sub-layer lower surface **542** may form at least part of the lower coating surface **512** of the coating **500**. The second sub-layer upper surface **551** may form at least part of the upper major surface **11** of the building panel **10**.

The first sub-layer **540** may form a physical barrier that at least partially separates the body **100** from the second sub-layer **550**. The physical barrier formed by the first sub-layer **540** may prevent at least some of the second sub-layer **550** (which comprises the glass-forming composition) from penetrating into the body **100** (e.g., a cellulosic body **100** having porous surfaces). According to some embodiments, the phosphoborate glass heat-insulative barrier that is created by glass-forming composition of the second sub-layer **540** may be separated from at least one of the first major surface **111**, second major surface **112**, and/or side surface **113** of the body **100** by a distance equal to the first sub-layer thickness t_{C1} .

The third sub-layer **560** may be formed by applying the moisture barrier composition with the addition of one or more organic solvents. Non-limiting examples of organic solvents include toluene, ethanol, acetone, butyl acetate, methyl ethyl ketone, ethyl 3-ethoxypropionate. The barrier composition may be present relative to the organic solvent in a weight ratio ranging from about 5:1 to about 1:20. After application to the second sub-layer upper surface **551**, the moisture barrier composition may be dried for a third period of time, optionally at an elevated temperature, sufficient to drive off any organic solvent. The resulting third sub-layer **560** may be a continuous or discontinuous coating having an third sub-layer upper surface **561** and a third sub-layer lower surface **562** opposite the third sub-layer upper surface **561**.

In the wet-state, the third sub-layer **560** may be applied in an amount such that the third sub-layer **560** has a wet thickness ranging from about 1.0 mils to about 10.0 mils—including all values and sub-ranges there-between. After drying, the third sub-layer **560** in the dry-state may have a third sub-layer thickness “ t_{C3} ” as measured from the third sub-layer upper surface **561** to the third sub-layer lower surface **562**. The third sub-layer thickness t_{C3} may range from about 1 mils to about 6 mils. The third sub-layer **560** may comprise a third sub-layer side surface **563** that extends from the third sub-layer upper surface **561** to the third sub-layer lower surface **562** and forms a perimeter of the second sub-layer **560**.

According to such embodiments, the overall coating thickness t_C of coating **500** may be the summation of the first sub-layer thickness t_{C1} , the second sub-layer thickness t_{C2} , and the third sub-layer thickness t_{C3} —as follows:

$$t_C = t_{C1} + t_{C2} + t_{C3}$$

According to these other embodiments, the first sub-layer lower surface **542** of the first sub-layer **540** may contact the upper cellulosic surface **411** of the cellulosic layer **400**. The first sub-layer upper surface **541** may contact the second sub-layer lower surface **552** of the second sub-layer **550**. The second sub-layer upper surface **551** may contact the third sub-layer lower surface **562** of the second sub-layer **560**. The third sub-layer upper surface **561** may form at least part of the upper coating **511** of the coating **500**. The first sub-layer lower surface **542** may form at least part of the lower coating surface **512** of the coating **500**. The third sub-layer upper surface **561** may form at least part of the upper major surface **11** of the building panel **10**.

According to other embodiments, the coating **500** may comprise only the second sub-layer **550** and the third sub-layer **560** without the first sub-layer **540** (not pictured). In such embodiments, the second sub-layer **550** may be directly atop one of the first major surface **111**, second major surface **112**, and/or side surface **113** of the body **100** and the third sub-layer **560** may be directly atop the second sub-layer upper surface **551** of the second sub-layer **550**. In such embodiments, the second sub-layer **550** acts as a sealant and is capable of sealing the one of the first major surface **111**, second major surface **112**, and/or side surface **113** of the body **100**, while simultaneously acting as an insulative barrier created from the glass-forming composition of the inorganic composition.

According to other embodiments, the coating **500** may comprise only the second sub-layer **550**. In such embodiments, the second sub-layer **550** acts as a sealant and is capable of sealing one of the first major surface **111**, second major surface **112**, and/or side surface **113** of the body **100**, while simultaneously acting as an insulative barrier created from the glass-forming composition of the inorganic composition, in situations where moisture resistance of the coating is not required.

Referring now to FIGS. 2-4, the body **100** may comprise a cellulosic layer **400**, whereby the coating **500** is applied directly to the cellulosic layer **400** of the body **100**. In other embodiments where the body **100** may be a laminate structure comprising multiple layers that includes a cellulosic layer **400** atop a substrate layer **200** with an adhesive layer **300** positioned there-between. The body **100** may also comprise the substrate layer **200** without the cellulosic layer **400** or adhesive layer **200**, whereby the coating **500** is applied to at least one surface of the substrate layer **200**—as discussed further herein.

Referring to FIG. 2, the cellulosic layer **400** may comprise an upper cellulosic surface **411** and a lower cellulosic surface **412** opposite the upper cellulosic surface **411**. The cellulosic layer **400** may comprise a cellulosic side surface **413** that extends from the upper cellulosic surface **411** to the lower cellulosic surface **412** and forms a perimeter of the cellulosic layer **400**. The cellulosic side surface **413** may form a portion of the side surface **113** of the body **100**. Stated otherwise, the side surface **113** of the body **100** may comprise the cellulosic side surface **413**. The side surface **113** of the body **100** may form a major side surface **13** of the building panel **10**. The first major surface **111** of the body **100** may comprise the upper cellulosic surface **411**.

The cellulosic layer **400** may have a cellulosic layer thickness " t_{CL} " as measured by the distance between the upper and lower cellulosic surfaces **411**, **412**. The cellulosic layer thickness t_{CL} may range from about 10 mils to about 3,000 mils—including all values and sub-ranges there-between. In some embodiments, the cellulosic layer **400** may form a veneer that is bonded to the substrate layer **200**

by the adhesive layer **300**, whereby the cellulosic thickness t_{CL} may range from about 10 mils to about 100 mils—including all values and sub-ranges there-between. In other embodiments, the cellulosic layer **400** may form the entirety of the body **100**, whereby the cellulosic thickness t_{CL} may range from about 300 mils to about 3,000 mils—including all values and sub-ranges there-between.

The cellulosic layer **400** may be formed from a cellulosic material such as wood, bamboo, and a combination thereof, and may be naturally occurring or engineered. Non-limiting examples of wood include cherry, maple, oak, walnut, pine, poplar, spruce, chestnut, mahogany, rosewood, teak, ash, hickory, beech, birch, cedar, fir, hemlock, basswood, alder wood, obeche wood, and combinations thereof. The cellulosic layer **400** may comprise pores that are not only present within the body of the cellulosic layer **400** but also exposed on at least one of the upper cellulosic surface **411**, lower cellulosic surface **412**, and/or the cellulosic side surface **413**. The porosity of the cellulosic layer **400** will depend on the bamboo or type of wood selected as the material that forms the cellulosic layer **400**.

The benefit of using a cellulosic layer **400** is that the resulting building panel **10** will exhibit authentic decorative features **30** of real wood and/or bamboo (e.g., wood grain, knots, burl, etc.) while minimizing the overall thickness required for the building panel **10** without necessitating artificial print layers. Artificial print layers, such as those on various papers or plastics, have been used as a way to recreate wood grain, knots, burl, etc., while minimizing layer thickness. Such print layers, however, are undesirable because of the limited amount of variation the cellulosic pattern across a large number of panels as compared to the same large number of panels that use cellulosic layers formed from real wood and/or bamboo. Stated otherwise, artificial print layers are not preferred because of the repetition in the decorative pattern over large installation areas.

Although not limited to this embodiment, the coating **500** may be directly atop the upper cellulosic surface **411**, the lower cellulosic surface **412**, and/or the cellulosic side surface **413** of the cellulosic layer **400**. The coating **500** may be applied to the cellulosic layer **400** such that the lower coating surface **512** is in direct contact with the upper cellulosic layer surface **411**. In such embodiments, the lower coating surface **512** may directly contact the upper cellulosic surface **411**, such that the upper coating surface **511** forms at least a portion of the upper major surface **11** of the building panel **10**.

Referring now to FIG. 2, according to some embodiments of the present invention the building panel **10** may include cellulosic layer **400** being adhesively bonded to the substrate layer **200** by an adhesive layer **300**. The combination of layers **200**, **300**, **400**, **500** of the present invention creates a laminate structure having high lamination integrity in a ceiling system under both standard conditions (i.e. daily operation of an interior building environment) but also during exposure to the extreme heat and temperature that may result from a fire.

The substrate layer **200** may comprises an upper substrate surface **211** and a lower substrate surface **212** that is opposite the upper substrate surface **211**. The substrate layer **200** may comprise a substrate side surface **213** that extends from the upper substrate surface **211** to the lower substrate surface **212** and forms a perimeter of the substrate layer **200**. The substrate side surface **213** may form a portion of the side surface **113** of the body **100**. Stated otherwise, the side surface **113** of the body **100** may comprise the substrate side surface **213**.

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The substrate layer **200** may be formed from a metallic material, ceramic material, or composite material. Non-limiting examples of metallic material include aluminum, steel, and iron. In a preferred embodiment, the substrate layer **200** is formed from aluminum. The substrate layer **200** may have a substrate thickness " t_s " ranging from about 20 mils to about 100 mils—including all values and sub-ranges there-between. The substrate thickness t_s may range from about 25 mils to about 80 mils. In a preferred embodiment, the substrate thickness t_s ranges from about 30 mils to about 65 mils—including all values and sub-ranges there-between.

The adhesive layer **300** may comprise an upper adhesive surface **311** and a lower adhesive surface **312** opposite the upper adhesive surface **311**. The adhesive layer **300** may comprise an adhesive side surface **313** that extends from the upper adhesive surface **311** to the lower adhesive surface **312** and forms a perimeter of the adhesive layer **300**. The adhesive side surface **313** may form a portion of the side surface **113** of the body **100**. Stated otherwise, the side surface **113** of the body **100** may comprise the adhesive side surface **213**. The adhesive layer **300** may have an adhesive thickness " t_a " ranging from about 2 mils to about 20 mils—including all values and sub-ranges there-between—as measured from the upper adhesive surface **311** to the lower adhesive surface **312**. In a preferred embodiment, the adhesive thickness t_a ranges from about 5 mils to about 15 mils—including all values and sub-ranges there-between.

According to embodiments where the building panel **10** has the laminate structure, the overall panel thickness t_p of the building panel **10** may be the summation of the substrate thickness t_s , the adhesive thickness t_a , the cellulosic layer thickness t_{CL} , and the coating thickness t_C as follows:

$$t_p = t_s + t_a + t_{CL} + t_C$$

The upper substrate surface **211** of the substrate layer **200** may directly contact the lower adhesive surface **312** of the adhesive layer **300** and the upper adhesive surface **311** of the adhesive layer **300** may directly contact the lower cellulosic surface **412** of the cellulosic layer **400** such that the adhesive layer **300** adhesively bonds together the cellulosic layer **400** and the substrate layer **200**. The lower coating surface **512** may directly contact the upper cellulosic surface **411**, such that the upper coating surface **511** forms at least a portion of the upper major surface **11** of the building panel **10**. In such embodiments, the lower substrate surface **212** may form at least a portion of the lower major surface **12** of the building panel **10**.

The adhesive layer **300** may be formed from an adhesive composition that is a hot-melt composition, water-based polyvinyl acetate adhesive, and combinations thereof. According to the purposes of the present invention, the term "hot-melt adhesive composition" means a composition having a melt viscosity that ranges from about 10,000 centipoise to about 40,000 centipoise at a temperature of about 275° F.—including all values and sub-ranges there-between. The hot-melt adhesive composition may be solid at room temperature and be substantially free of solvent. The adhesive composition may comprise adhesive polymer in an amount ranging from about 50 wt. % to about 100 wt. % based on the total weight of the adhesive composition—including all values and sub-ranges there-between.

The adhesive polymer according to the present invention may be a thermoplastic polymer. Non-limiting examples of the thermoplastic polymer may include moisture cured poly-

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ester modified polyurethane polymers. Such polyester modified polyurethanes may be formed by reacting organic diisocyanate with difunctional polyester polyol and low molecular weight diols (as chain-extending agents) at a non-limiting NCO:OH ratio of about 0.7:1 to about 1.3:1—including all sub-ranges and ratios there-between.

Non-limiting examples of polyester polyol include difunctional polyester diols containing alcoholic hydroxyl groups. Suitable polyester diols are polyester having average molecular weights of from 800 to 5000 and preferably from 2000 to 4000 produced from (i) dicarboxylic acids containing at least 6 carbon atoms, such as adipic acid, pimelic acid, suberic acid, azelaic acid and/or sebacic acid (preferably adipic acid, as the sole acid component) and (ii) alkane diols that may contain at least 4 carbon atoms, such as, for example, 1,4-dihydroxy-butane, 1,5-dihydroxypentane and/or 1,6-dihydroxy-hexane. Polycondensates of ω -hydroxyalkane-mono-carboxylic acids and the polymers of their lactones are also suitable, although less preferred.

Low molecular weight diols suitable as chain-extending agents in accordance with the present invention include, in particular, aliphatic diols having average molecular weight of from 62 to 400 or mixtures thereof. Non-limiting examples of such diols include ethylene glycol, 1,3-dihydroxy-propane, 1,4-dihydroxy-butane, 1,5-dihydroxypentane, 1,6-dihydroxyhexane, and the like.

Non-limiting examples of suitable aromatic polyisocyanates include all isomers of toluylene-diisocyanate (TDI), naphthalene-1,5-diisocyanate, diphenylmethane-4,4'-diisocyanate (MDI), diphenylmethane-2,4'-diisocyanate and mixtures of 4,4'-diphenylmethane-diisocyanate with the 2,4' isomer or mixtures thereof with oligomers of higher functionality (so-called crude MDI), xylylene-diisocyanate (XDI), 4,4'-diphenyl-dimethylmethane-diisocyanate, di- and tetra-alkyl-diphenylmethane-diisocyanate, 4,4'-dibenzyl-diisocyanate, 1,3-phenylene-diisocyanate and 1,4-phenylene-diisocyanate. Examples of suitable cycloaliphatic polyisocyanates are the hydrogenation products of the above-mentioned aromatic diisocyanates, such as 4,4'-dicyclohexylmethane-diisocyanate (H_{12} MDI), 1-isocyanatomethyl-3-isocyanato-1,5,5-trimethyl-cyclohexane (isophorone-diisocyanate, IPDI), cyclohexane-1,4-diisocyanate, hydrogenated xylylene-diisocyanate (H_6 XDI), 1-methyl-2,4-diisocyanato-cyclohexane, m- or p-tetramethylxylene-diisocyanate (m-TMXDI, p-TMXDI) and dimer-fatty acid diisocyanate. Examples of aliphatic polyisocyanates are tetramethoxybutane-1,4-diisocyanate, butane-1,4-diisocyanate, hexane-1,6-diisocyanate (HDI), 1,6-diisocyanato-2,2,4-trimethylhexane, 1,6-diisocyanato-2,4,4-trimethylhexane and 1,2-dodecane-diisocyanate (C_{12} DI).

The adhesive composition of the present invention may further comprise additives selected from the group consisting of 2,2'-dimorpholinethyl ether catalyst, di(2,6-dimethylmorpholinoethyl)ether catalyst, adhesion promoters, diluents, plasticizers, fillers, antioxidants pigments, UV absorbers and combinations thereof. In other embodiments, the adhesive composition may further comprise a flame retardant. Non-limiting examples of flame retardant may include ammonium hydroxide, magnesium hydroxide, huntite, hydromagnesite, silica, polyphosphate, melamine cyanurate, chloride salts—such as sodium chloride, antimony oxide, and borates, such as calcium borate, magnesium borate, zinc borate, and combinations thereof. The flame retardant may be present in the adhesive composition in an amount ranging from about 0 wt. % to about 50 wt. % based on the total weight of the adhesive composition—including all values and sub-ranged there-between.

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The building panel **10** of FIG. **2** may be formed heating the adhesive composition to an application temperature ranging from about 245° F. to about 330° F., and applying the adhesive composition to at least one of the upper substrate surface **211** or the lower cellulosic surface **412**. The adhesive composition may be applied by roll coating, spray coating, dip coating, or the like. Within the open time of the adhesive (typically 30 to 60 seconds), the upper substrate surface **211** is mated to the lower cellulosic surface **412** with the adhesive composition being present therebetween, thereby bonding the upper substrate surface **211** to the lower cellulosic surface **412** via the adhesive composition. Pressure may then be applied to at least one of the upper cellulosic surface **411** of the cellulosic layer **400** or the lower substrate surface **212** of the substrate layer **200** to ensure proper adhesive bonding.

Each sub-layer **540**, **550**, **560** may be individually applied by spray, roll-coating, dip coating, curtain coating, brushing, blade coating, or the like. Specifically, the first sub-layer **540** may be applied to the upper cellulosic surface **411** of the cellulosic layer **400**. The first sub-layer **540** may then be optionally heated to a temperature ranging from about 68° F. to about 150° F. to partially or fully cure the first sub-layer **540**. The second sub-layer **550** may then be applied to the first sub-layer upper surface upper surface **541**. The second sub-layer **550** may then be optionally heated to a temperature ranging from about 190° F. to about 320° F. to partially or fully cure the second sub-layer **550**. The third sub-layer **560** may then be applied to the second sub-layer upper surface **551**. The third sub-layer **560** may then be optionally heated to a temperature ranging from about 190° F. to about 320° F. to partially or fully cure the third sub-layer **560**—thereby resulting in the laminate structure of the present invention. The laminate structure may then be heated in an oven to fully cure the adhesive layer **300** and the coating **500** for a fourth period of time.

According to the present invention, the coating **500** applied to the cellulosic layer **400** provides an aesthetically pleasing building panel **10** such that the underlying body decorative features on the body **100** are visible from the upper major surface **11** of the building panel **10** as the decorative features **30**, because the decorative coating **500** is substantially clear. Furthermore, the inorganic composition of the coating **500** helps provide an insulative heat-barrier to the cellulosic layer **400**, thereby helping prevent the cellulosic layer **400** from igniting during a fire and propagating through the building panel **10**.

The multi-layered coating **500** comprising the first sub-layer **540** may also at least partially seal the pores and the upper cellulosic surface **411** such that at least a portion of the phosphoborate glass layer is formed at a distance separated from the upper cellulosic surface **411** of the cellulosic layer **400**—further protecting the cellulosic layer **400** from igniting in a fire. Additionally, the moisture sealant composition of the third sub-layer **560** ensures that the glass-forming composition of the underlying sub-layers **540**, **550** remains active for prolonged periods of time in case an interior space catches fire years after initial installation.

Referring to FIG. **5**, the building panel **10** of the present invention may be a ceiling panel (as shown installed in the ceiling system of FIG. **5**), a wall panel, or the like. The lower major surface **12** of the ceiling panel **10** of the present invention may face the plenum space **3** of an interior space of a ceiling system **1**. The upper major surface **11** of the ceiling panel **10** of the present invention may face the active space **2** of an interior space of a ceiling system **1**.

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In non-exemplified embodiments, the present invention may include a building panel having an upper major surface opposite a lower major surface, the building panel comprising a cellulosic layer (also referred to as “cellulosic body” in this embodiment) and a coating. The cellulosic body is self-supporting and comprises an upper cellulosic surface and a lower cellulosic surface opposite the upper cellulosic body. Non-limiting examples of a cellulosic body may include MDF board, wooden planks, or the like. The cellulosic body may have a cellulosic body thickness as measured from the lower cellulosic surface to the upper cellulosic surface that ranges up to about 3 inches—including all values and sub-ranges there-between.

With the coating **500** being formed at drying temperatures as low as 150° F., the cellulosic body may at least partially retain pre-existing moisture already contained within the cellulosic body. The surprising benefit of retaining the pre-existing moisture is that during exposure to high-heat, the retained moisture is converted to steam and driven out of the cellulosic body. As the steam escapes from the body **100**, the glass layer formed from the coating **500** is pushed outward from the body **100**, thereby increasing the distance between the body **100** and the surrounding flame or high-heat—thereby decreasing the likelihood that the body **100** ignites. Stated otherwise, it has been surprisingly discovered that the coatings **500** of the present invention further enhance fire repellency in the building panels **10** by allowing for drying temperatures below 212° F. under atmospheric conditions (at 1 atm).

The building panel **10** of such embodiments may have the coating **500** applied to at least one of the upper cellulosic surface **411** or the lower cellulosic surface **412** of the cellulosic body **400**. The coating **500** comprises an upper coating surface **511** opposite a lower coating surface **512**, whereby the lower coating surface **512** of the coating **500** may directly contact the upper cellulosic surface **411** of the cellulosic body **400**. The coating **500** comprises at least the second sub-layer **550** and optionally the first sub-layer **540** and/or the third sub-layer **560**, as previously discussed. The upper major surface **11** of the building panel **10** may comprise the upper coating surface **511** of the coating **500**. According to some embodiments, the lower major surface **12** of the building panel **10** may be uncoated, whereby the lower major surface **12** of the building panel **10** does not comprise the coating **500**, but rather the lower cellulosic surface **412** or the lower substrate surface **212**.

Referring now to FIGS. **6-8**, a building panel **1010** and ceiling system **1001** are illustrated in accordance with another embodiment of the present invention. The building panel **1010** is similar to the building panel **10** except as described herein below. The description of the building panel **10** above generally applies to the building panel **1010** described below except with regard to the differences specifically noted below. A similar numbering scheme will be used for the building panel **1010** as with the building panel **10** except that the 1000-series of numbers will be used. Additionally, the ceiling system **1001** is similar to the ceiling system **1** except as described herein below. The description of the ceiling system **1** above generally applies to the ceiling system **1001** described below except with regard to the differences specifically noted below. A similar numbering scheme will be used for the ceiling system **1001** as with the ceiling system **1** except that the 1000-series of numbers will be used.

The coating **1500** of the present invention may be applied to one of the first major surface **1111**, second major surface **1112**, and/or side surface **1113** of the body **1100**. In a

preferred embodiment, the coating **1500** is applied to each of the first major surface **1111**, the second major surface **1112**, and the side surface **1113** of the body **1100** such that the coating **1500** encapsulates the body **1100**. Stated otherwise, the coating **1500** may form a continuous barrier that substantially surrounds the entire body **1100**.

Although not pictured, the coating **1500** of this embodiment comprises the second sub-layer **550**, and may further comprise each of the first sub-layer **540** and/or third sub-layer **560**, as previously discussed with respect to FIGS. 2-4.

In these embodiments, the body **1100** may be formed entirely from the cellulosic body **400** or the substrate layer **200**. The body **1100** may alternatively be formed from the laminate structure. According to the embodiments where the body **1100** is formed entirely from the cellulosic body **400**, the overall panel thickness t_P may be the summation of cellulosic body thickness t_{CL} , and the coating thickness t_C as follows:

$$t_P = t_{CL} + [n \times t_C]$$

whereby the “n” refers to the number of major surfaces of the body **1100** coated with the coating **1500**. In this embodiment, the first and second major surfaces **1111**, **1112** of the body **1100** are coated and $n=2$. In such embodiments, the panel thickness t_P may range from about 500 mils to about 2,000 mils—including all values and sub-ranges there-between.

According to the embodiments where the body **1100** is formed entirely from the substrate layer **200**, the overall panel thickness t_P may be the summation of substrate layer thickness t_S , and the coating thickness t_C as follows:

$$t_P = t_S + [n \times t_C]$$

whereby the “n” refers to the number of major surfaces of the body **1100** coated with the coating **1500**. In this embodiment, the first and second major surfaces **1111**, **1112** of the body **1100** are coated and $n=2$. In such embodiments, the panel thickness t_P may range from about 100 mils to about 600 mils—including all values and sub-ranges there-between.

Referring now to FIG. 8, the building panel **1010** of these embodiments may be installed into a ceiling system **1001** comprising a support **1005** that includes a first support

member **1005a** and a second support member **1005b**. The first and second support members **1005a**, **1005b** may be arranged in an intersecting pattern to form a support grid. A plurality of the building panels **1010** may be arranged in an array and attached to the support grid such that the upper major surface of one building panel **1010** faces the lower major surface of a second building panel **1010** that is adjacent to the first building panel **1010**. The plurality of the building panels **1010** may also be arranged and attached to the support grid such that the plurality of building panels **1010** comprises a first side **1041** opposite a second side **1042**, whereby the first side **1041** faces the plenum **1003** and the second side **1042** faces the room environment **1002**—the first and second sides **1041**, **1042** comprises the side surface **1013** of the building panels **1010**.

The following examples are prepared in accordance with the present invention. The present invention is not limited to the examples described herein.

EXAMPLES

Experiment 1

A first set of experiments was performed by coating a number of cellulosic bodies with the inorganic composition of the present invention in the wet-state. Specifically, a major surface of each cellulosic body had the dimensions of 3.75"×29.875", which were coated by the coating of the present invention. Each coated body was then dried at a temperature ranging between 200° F. to 275° F. Each specimen was then positioned above a Bunsen burner angled at 30°, whereby the coated major surface faces the flame from the Bunsen burner. Each surface was exposure for a set predetermined amount of time, after which the amount of flame spread on each specimen was measured and assigned a value—the lower the Flame Spread Rating (“FSR”) value, the better the coating was at imparting flame-retardancy to the underlying substrate.

Additionally, using a correlation between measured FSR values and known flame spread and smoke index values generated by an ASTM E-84 test, the FSR values were able to be correlated to Class A, B, or C flame retardancy. Specifically, FSR values <25 translate to high probability of a Class A fire rating. FSR values between 25 and 40 translate to a moderate probability of Class A fire rating. FSR values >40 translate to a low probability of Class A fire rating.

TABLE 1

	Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Ex. 9	Ex. 10	Ex. 11	Ex. 12	Ex. 13
Component (wt. %)	14A	14B	14C	23B	32A	50D	50E	103B	103C	107F	115A	115C	131B
Water	77.3	71.3	67.2	68.6	53.9	68.4	68.1	65.8	70.4	63.7	64.7	66.0	45.1
H ₃ PO ₄	13.6	17.2	20.4	20.9	32.3	20.7	20.7	23.0	17.0	21.3	21.5	22.0	35.7
HNO ₃	—	—	—	—	—	0.2	0.2	—	—	—	—	—	—
B ₂ O ₃	1.8	1.6	1.5	—	1.6	0.3	0.3	—	—	—	—	—	2.0
2ZnO•3 B ₂ O ₃ •3.5H ₂ O	—	—	—	—	—	—	—	3.3	3.2	2.8	2.9	4.4	—
CaCO ₃	—	—	—	—	—	—	0.3	—	—	0.6	0.6	0.6	0.9
Al ₂ (CO ₃) ₃	—	—	—	—	—	—	—	—	—	2.8	—	—	4.8
Li ₂ CO ₃	—	—	—	3.1	6.1	—	—	—	—	—	—	—	—
MgCO ₃	7.3	9.9	10.9	7.4	6.1	10.4	10.4	6.6	6.5	5.7	8.6	4.7	7.6
Fumed SiO ₂	—	—	—	—	—	—	—	1.3	—	1.4	—	0.6	1.0

TABLE 1-continued

Component (wt. %)	Ex. 1 14A	Ex. 2 14B	Ex. 3 14C	Ex. 4 23B	Ex. 5 32A	Ex. 6 50D	Ex. 7 50E	Ex. 8 103B	Ex. 9 103C	Ex. 10 107F	Ex. 11 115A	Ex. 12 115C	Ex. 13 131B
Hydroxyethylcellulose	—	—	—	—	—	—	—	—	2.9	—	—	—	—
Triethanolamine	—	—	—	—	—	—	—	—	—	1.7	1.7	1.7	2.9
Total	100	100	100	100	100	100	100	100	100	100	100	100	100
Total Amount of Organic	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.0	2.9	1.7	1.7	1.7	2.9
pH	3.5	3.83	3.47	3.90	2.26	2.87	2.65	2.51	2.51	2.46	2.50	2.50	2.10

With the formulations set forth in Table 1, the following heat and flame performance are provided below in Table 2.

TABLE 2

	Ex. 7	Ex. 11	Ex. 12	Ex. 13	Organic Sealer
Wet Density (g/ml)	1.23	1.24	—	1.29	1.02
Wet Thickness (mil)	0.2	0.35	0.35	0.5	1.6
Dry Thickness (mil)	0.05	0.12	0.12	0.28	0.50
Total Amount of Organic (Dry)	0.2	1.7	1.7	2.9	100%
FSR	37	87	87	29	Very Poor
Optical Clarity	—	78%	—	100%	108%
Gloss (60°)	—	59.3	—	51.5	25.0

As shown in Table 2, the coating comprising the inorganic composition of the present invention comprising minor amounts of organic composition (i.e., <1.0 wt. % based on total dry-weight) resulted in superior FSR of 37, as compared to Examples 11 and 12 having greater amounts of organic composition (>4 wt. %), which resulted in an FSR of 87—even at small application amounts (dry thicknesses as low as 0.05 mils). Additionally, each inorganic coating compositions of the present invention allows for superior flame resistance while still maintaining high visual clarity (i.e., >70%).

Furthermore, as demonstrated by Example 13, it has been surprisingly discovered that greater amounts of organic composition (i.e., about 5 wt. % based on the dry-weight) may be added to the inorganic composition while still maintaining desirable FSR performance when the inorganic composition further comprises aluminum carbonate $Al_2(CO_3)_3$.

Furthermore, while the gloss value of the present invention may increase relative to the organic coating, such values still remain within an acceptable range of increase of such values compared to the organic coating. More importantly, the coatings of the present invention provide coatings having superior changes in optical clarity. According to the present invention, the percent change in optical clarity is a measure of color values before and after application of the coating, whereby when the color values improve (i.e., increase) there is a corresponding increase in optical clarity as the appearance of the underlying substrate has become more apparent. Thus, achieving >100% optical clarity is possible as it represents an improvement in discernible color of the underlying substrate through the respective coating.

Experiment 2

A second set of experiments was performed by coating a number of cellulosic bodies with a multi-layered coating comprising the inorganic composition of the present invention in the wet-state. Each building panel was then subjected

to color and gloss measurements as well as pencil hardness tests. The results are provided below in Table 3.

TABLE 3

	Phosphate Coating System	Control	Untreated Wood
Base Coat	Organic	Organic	None
Intermediate Coat	Ex. 13	Flame-Retardant Organic	None
Topcoat	Ex. 11	Flame-Retardant Organic	None
Gloss (60°)	19.4	21	3.2
Color			
L	65.4	80.4	71.95
a	7.16	2.7	3.85
b	23.92	23	20.22
Pencil Hardness	8H	2H	None

As demonstrated by Table 3, the multi-layered coating comprising the inorganic composition of the present invention (referred to as “Phosphate Coating System”) demonstrates desirable color and gloss performance when compared to the flame-retardant organic coating. Specifically, the Phosphate Coating System exhibits slight darkening (as measured by the decreased “L” value) as well as an increase in red color (as measured by the increase in “a” value”) however, such values remain within an acceptable range of optical performance—especially when considering the drastic improvement in flame retardancy of the Phosphate Coating System compared to the Flame-Retardant Organic coating—as demonstrated in Experiment 3.

Experiment 3

A third experiment was performed by subjecting each of the panels of Experiment 2 to the FSR test. The results are provided below in Table 4.

TABLE 4

	Phosphate Coating System	Control	Untreated Wood
Base Coat	Organic Coating	Organic Coating	None
Intermediate Coat	Ex. 13	Flame-Retardant Organic	None
Topcoat	Ex. 11	Flame-Retardant Organic	None
FSR	29	90-97	99

As demonstrated by Table 4, the multi-layered coating of the present invention resulted in superior FSR performance as compared to the organic based multi-layered coating of the Control.

What is claimed is:

1. A flame-retardant coating composition comprising:
a coating comprising a first sub-layer, a second sub-layer
atop the first sub-layer, and a third sub-layer atop the
second sub-layer;
wherein the first sub-layer is an organic coating and
comprises polymer produced from unsaturated mono-
mers and the second sub-layer is an inorganic coating
comprising aluminum carbonate, triethanolamine, and
a blend of a first borate compound and a first phosphate
compound;
wherein the aluminum carbonate is present in an amount
of about 2.8 to about 4.8 wt. %;
wherein a weight ratio of first phosphate compound to the
first borate compound within the second sub-layer is
from about 5:1 to about 80:1;
wherein the third sub-layer is an inorganic coating com-
position comprising triethanolamine, a second borate
compound, and a second phosphate compound; and
wherein the second borate compound is zinc borate and
the first borate compound is boron trioxide.
2. The flame-retardant coating of claim 1, wherein the
flame-retardant coating composition is optically transparent.
3. The flame-retardant coating of claim 1, wherein the
inorganic coating has a maximum pH of about 5.
4. The flame-retardant coating of claim 1, wherein the
inorganic coating comprises up to about 5 wt. % of organic
components based on the total weight of the inorganic
coating.
5. The flame-retardant coating of claim 1, wherein the first
borate compound is present in an amount of about 0.5 wt. %
to about 15.0 wt. %, based on the total dry weight of the
second sub-layer.
6. The flame-retardant coating of claim 1, wherein the first
sub-layer comprises an organic component selected from the
group consisting of a wax emulsion, latex polymer, and
combinations thereof.
7. The flame-retardant coating of claim 1, wherein the
second sub-layer and the third sub-layer comprise calcium
carbonate.
8. The composition of claim 1, wherein the first phosphate
compound is present in an amount from about 55.0 wt. % to
about 75.0 wt. %, based on the total weight of the second
sub-layer in the dry-state, and wherein the first borate
compound is present in an amount of about 1 wt. % to about
10.5 wt. %, based on the total weight of the second sub-layer
in the dry-state.

9. The composition of claim 1, further comprising silicon
dioxide.
10. A flame-retardant coating composition comprising:
a coating comprising a first sub-layer, a second sub-layer
atop the first sub-layer, and a third sub-layer atop the
second sub-layer;
wherein the second sub-layer comprises an inorganic
component comprising aluminum carbonate, trietha-
nolamine, a filler comprising magnesium carbonate and
calcium carbonate, and a blend of a first borate com-
pound and a first phosphate compound comprising
phosphoric acid;
wherein the third sub-layer comprises a second inorganic
component comprising triethanolamine, magnesium
carbonate, and a blend of a second borate compound
that is different from the first borate compound and a
second phosphate compound;
wherein the aluminum carbonate is present in an amount
from about 2.8 to about 4.8 wt. %;
wherein the coating composition has a maximum pH of 5
and comprises up to about 5.0 wt. % of organic
components based on the total weight of the coating
composition;
wherein the composition is free of carrier;
wherein the first borate compound is boron trioxide and
the second borate compound is zinc borate;
wherein a weight ratio of phosphate to the borate com-
pound within the inorganic component of the second
sub-layer is from about 5:1 to about 80:1;
wherein the flame-retardant coating composition is opti-
cally transparent.
11. The composition of claim 10, wherein the trietha-
nolamine is present in an amount ranging from a non-zero
amount to less than 5.0 wt. %, based on the total weight of
the coating composition.
12. The composition of claim 10, wherein the composi-
tion comprises up to about 1.0 wt. % of organic components,
based on the total weight of the coating composition.
13. The composition of claim 10, wherein the first phos-
phate compound is present in an amount from about 55.0 wt.
% to about 75.0 wt. %, based on the total weight of the
coating composition, and wherein the first borate compound
is present in an amount of about 1 wt. % to about 10.5 wt.
%, based on the total weight of the coating composition.
14. The composition of claim 10, further comprising
silicon dioxide.

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